Paleomagnetism and U–Pb geochronology of diabase dyke swarms of Minto block, Superior Province, Quebec, Canada[1]

Kenneth L. Buchan, James K. Mortensen, Kenneth D. Card, and John A. Percival

Abstract: In the first collaborative study of paleomagnetism and precise U–Pb geochronology in the Minto block of the Superior Province, mafic dyke swarms with three widely divergent paleomagnetic signatures and isotopic ages have been identified. The 2505 ± 2 Ma Ptarmigan dykes trend north to northeast and have a virtual geomagnetic pole at 42°S, 220°E, similar to that of 2473–2446 Ma Matachewan dykes of the southern Superior Province. The ca. 2230 Ma Maguire dykes trend west to northwest and yield a paleopole at 9°S, 267°E, similar to those for 2216^{+8}_{-4} Ma Senneterre dykes and 2217–2210 Ma Nipissing sills of the southern Superior and Southern provinces, respectively. The 2209 ± 1 Ma Klotz dykes trend west-northwest, but do not carry a consistent magnetization direction. Finally, 1998 ± 2 Ma Minto dykes of west-northwest to northwest trend, identical in age to the 1998 Ma ± 2 Ma Purtuniq ophiolite of the Cape Smith Belt, have a paleopole at 38°N, 174°E. The similarity of paleopoles for the ca. 2.23–2.21 Ga Maguire dykes of the Minto block, Senneterre dykes of the southern Superior, and Nipissing sills of the Southern Province demonstrates that these regions were in their present relative latitudes and orientations at that time. Likewise, the similarity of the Ptarmigan virtual geomagnetic pole and the Matachewan paleopole suggests little relative latitudinal movement or rotation of the two regions since ca. 2.5 Ga. The Maguire, Senneterre, and Klotz dykes form a roughly radiating pattern and may represent one quadrant of a giant radiating dyke swarm centred southeast of Ungava Bay, whose focus marks the location of a mantle plume responsible for ca. 2.22 Ga breakup along the eastern margin of the Superior Province. If so, the coeval Nipissing sills that intrude sedimentary rocks of the Huronian Supergroup of the Southern Province may have been fed laterally by Senneterre dykes from the Ungava plume centre.

Résumé : Lors de la première étude en collaboration entreprise sur le paléomagnétisme et la géochronologie U–Pb précise du bloc de Minto de la Province du Supérieur, on a pu identifier des essaims de dykes mafiques caractérisés par trois signatures paléomagnétiques nettement divergentes et par des âges isotopiques différents. Les dykes de Ptarmigan datés de 2505 ± 2 Ma qui présentent une direction nord et nord-est possèdent un pôle géomagnétique virtuel à 42°S, 220°E, analogue à celui des dykes de Matachewan âgés de 2473–2446 Ma dans la partie sud de la Province du Supérieur. Les dykes de Maguire, vieux d'environ 2230 Ma, affichent une direction ouest et nord-ouest et fournissent un paléopôle à 9°S, 267°E, analogues aux dykes de Senneterre datés de 2216^{+8}_{-4} Ma et aux filons-couches de Nipissing âgés de 2217–2210 Ma dans le sud de la Province du Supérieur et de la Province du Sud, respectivement. Les dykes de Klotz datés de 2209 ± 1 Ma sont orientés ouest-nord-ouest, mais sans direction d'aimantation uniforme. Finalement, les dykes de Minto âgés de 1998 ± 2 Ma, orientés ouest-nord-ouest à nord-ouest, ont le même âge que l'ophiolite de Purtuniq (1998 ± 2 Ma) de la Zone de Cape Smith, et leur paléopôle est à 38°N, 174°E. La concordance qui existe entre les paléopôles des dykes de Maguire du bloc de Minto datés d'environ 2,23–2,21 Ga, des dykes de Senneterre de la région sud de la Province du Supérieur et des filons-couches de Nipissing de la Province du Sud démontre que ces régions occupaient toutes en ces temps les mêmes latitudes et qu'elles avaient les mêmes orientations qu'aujourd'hui. De même, la coïncidence du pôle géomagnétique virtuel de Ptarmigan avec le paléopôle de Matachewan suggère un très faible déplacement latitudinal relatif ou de rotation des deux régions depuis 2,5 Ga. Les dykes de Maguire, Senneterre et Klotz, formant un arrangement approximativement radiant, peuvent représenter un quadrant d'un très grand essaim de dykes radiants centré au sud-est de la baie d'Ungava, et dont le foyer marquerait la localisation du panache mantellique responsable de la fragmentation survenue vers 2,22 Ga, le long de la marge orientale de la Province du Supérieur. Si cette interprétation est juste, les filons-couches de Nipissing contemporains qui pénètrent les roches sédimentaires du Supergroupe de l'Huronien de la Province du Sud ont pu être nourris latéralement par les dykes de Senneterre du centre du panache d'Ungava.

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K.L. Buchan[2] and J.A. Percival. Geological Survey of Canada, 601 Booth Street, Ottawa, ON K1A 0E8, Canada.
J.K. Mortensen. Department of Geological Sciences, University of British Columbia, 6339 Stores Road, Vancouver, BC V6T 1Z4, Canada.
K.D. Card. Card and Associates Geosearch, 86 Penfield Drive, Kanata, ON K2K 1M1, Canada.

[1]Geological Survey of Canada Contribution 1997223.
[2]Corresponding author (e-mail: kbuchan@NRCan.gc.ca).

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Introduction

Numerous diabase dyke swarms, mostly of Proterozoic age, intrude the Superior Province of the Canadian Shield and are thought to record periodic extensional magmatism linked to events at the margins of the craton. Recent collaborative studies of paleomagnetism and U–Pb geochronology of dyke swarms of the southern Superior Province have yielded precise ages for Paleoproterozoic paleomagnetic poles and have resulted in a major revision of the Paleoproterozoic apparent polar wander path for this area (Buchan et al. 1993, 1994, 1996). Using the same approach, we have carried out the first detailed study of Neoarchean and Paleoproterozoic diabase dykes of the Minto block in the northeastern Superior Province (Fig. 1). This study was designed to (1) establish the age, trend, and distribution of dyke swarms in the Minto block, (2) determine their relationship to other Paleoproterozoic magmatism and tectonic events in and around the Superior Province, (3) determine well-dated paleopoles for the Minto block in order to investigate its drift in the Paleoproterozoic, and (4) compare paleomagnetic poles of the Minto block and the southern Superior Province to test models explaining the prominent difference in structural trends between the two regions.

The Minto block is transected by undeformed dykes of at least five different trends (Fig. 1). However, with the exception of a preliminary paleomagnetic study (Fahrig et al. 1986) of five sampling sites in the north-northwest-trending Payne River swarm (Fig. 1), no paleomagnetism or U–Pb geochronology has been reported from these dykes.

The stably magnetized dykes from the Minto block in this study each exhibit one of three distinct directions of remanent magnetization, suggesting that dykes of three ages are present. A U/Pb age on a dyke from each directional grouping confirmed this interpretation. Ptarmigan dykes were dated at 2505 ± 2 Ma, Maguire dykes at 2229^{+35}_{-20} Ma, and Minto dykes at 1998 ± 2 Ma (Fig. 1). In addition, a dyke from the Klotz swarm, which does not carry a stable direction of magnetization, was dated at 2209 ± 1 Ma.

Geological setting

The Minto block of the northeastern Superior Province (Fig. 1) is characterized by northerly structural trends, which contrast easterly trends in the southern part of the province (inset to Fig. 1). It records a history extending from >3.10 to 2.65 Ga, with tectono-magmatic events including production of tonalitic crust at 3.10–2.90 Ga, eruption of juvenile submarine volcanic rocks, deformation, and emplacement of plutons at 2.84–2.77 Ga, and eruption of calc-alkaline volcanic rocks, deformation, and emplacement of charnockitic and hydrous plutons at 2.76–2.69 Ga (Percival et al. 1992, 1994; Stern et al. 1994; Skulski et al. 1996; Skulski and Percival 1996). The resultant geometry consists of narrow (1–10 km), elongate (40–100 km) amphibolite-facies metavolcanic–metasedimentary belts separated by magnetically high massifs of dominantly pyroxene-bearing granitoid rocks (charnockites; Percival et al. 1996). Events of similar character and age are common throughout the northern Superior Province including northwestern Ontario (Thurston et al. 1991) and reflect widespread crustal growth and recycling events. Percival et al. (1994) explained the prominent difference in structural trends in a common kinematic framework for the Superior Province at ca. 2.7 Ga. In this model, the Minto block formed as north-trending (present-day coordinates) continental magmatic arcs, built on the eastern margin of a ca. 3 Ga protocraton that is preserved as isolated remnants and geochemical inheritance in granitoid rocks of the western Minto block. The older core extends across Hudson Bay into the North Caribou Terrane of northwestern Ontario (Fig. 1 inset; Thurston et al. 1991), and acted as a buttress to northward accretion and related westward tectonic escape of oceanic and microcontinental terranes, now represented by the more southerly east–west-trending subprovinces of the western Superior Province.

Following the major tectonic activity, metamorphism, and plutonism at ca. 2.7 Ga, the Minto block cooled to record ca. 2.5 Ga K–Ar mica dates. It was transected by several swarms of diabase dykes (Fig. 1), some of which are likely linked to Paleoproterozoic rifting events (ca. 2.2–2.0 Ga) at the margins of the Superior Province. Only minor deformation and retrogression of the Superior craton is associated with subsequent (ca. 1.8 Ga) compressional orogeny in the New Quebec and Ungava orogens.

In this study, the Ptarmigan, Maguire, Klotz, and Minto dyke swarms (Fig. 1) are defined on the basis of their U–Pb geochronology, paleomagnetism, petrographic characteristics, and trend. The extent of Ptarmigan, Maguire, and Minto swarms is poorly defined because the dykes are not readily traceable on aeromagnetic maps. Dykes of the Klotz swarm are also not evident on aeromagnetic maps, but can be easily traced on aerial photographs.

The 2505 Ma Ptarmigan swarm consists of only a few dykes (Fig. 1). These brown-weathering, north- to northeast-trending dykes, up to 45 m wide, are characterized by coarse clinopyroxene (25%), orthopyroxene (20%), and olivine (5%) in a matrix of calcic plagioclase (45%) and accessory Fe–Ti oxides, biotite, and apatite. They are generally fresh with minor uralitic amphibole after pyroxene, chlorite, serpentine, and magnetite after olivine, and sparse serpentine and chlorite-filled veinlets.

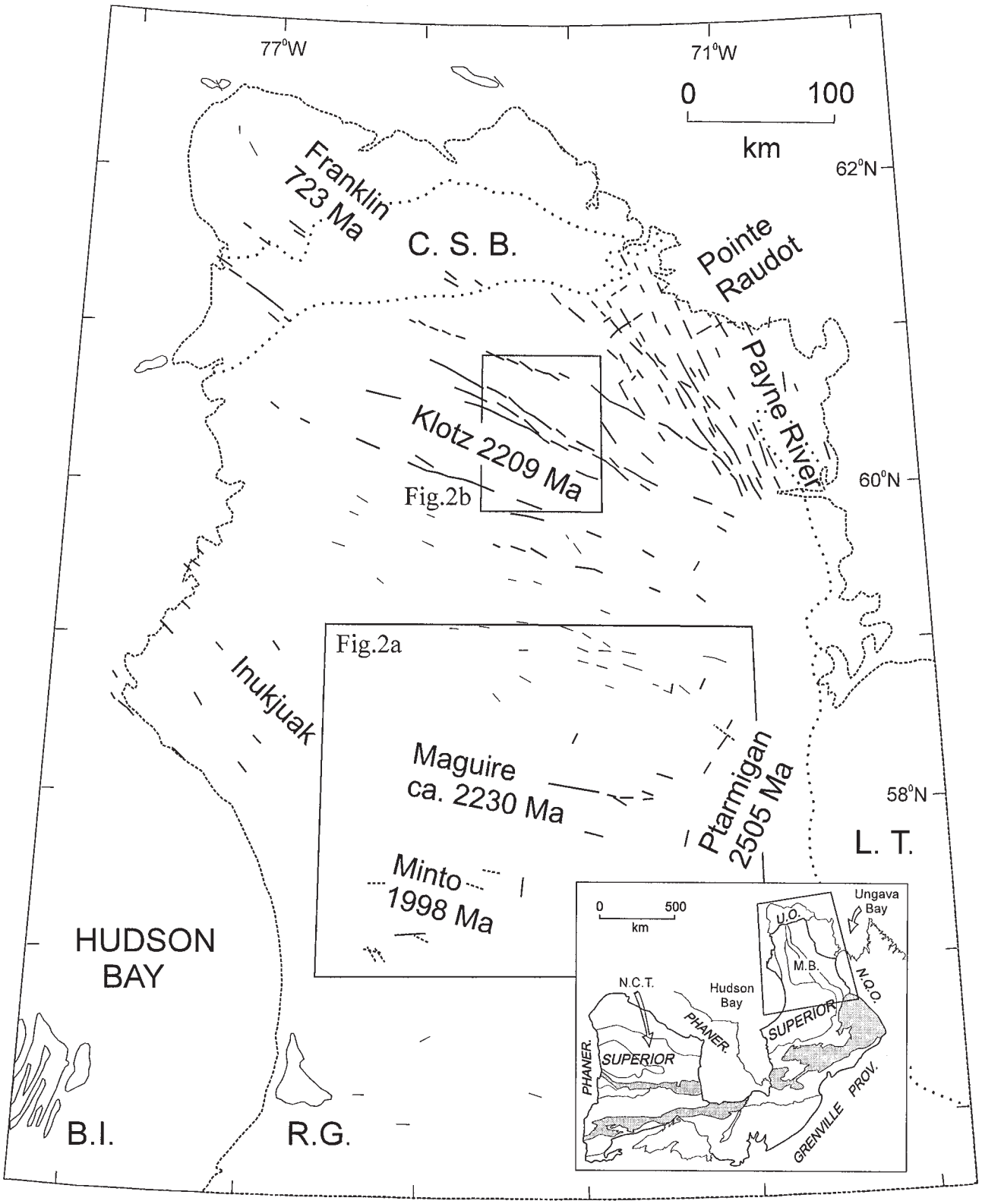
Dykes of the ca. 2230 Ma Maguire swarm trend west to northwest, weather green, and are markedly altered. These medium- to coarse-grained dykes, up to 45 m wide, consist of clinopyroxene (40%) and calcic plagioclase (50%), with minor quartz (5%), biotite, and Fe–Ti oxides (5%). Two petrographic types are distinguished: (1) those with subhedral to euhedral plagioclase surrounded by subhedral to anhedral clinopyroxene and (2) rocks with strongly zoned, subhedral to anhedral clinopyroxene in a matrix of saussuritic plagioclase.

The 2209 Ma Klotz swarm consists of dykes up to 100 m wide that form prominent west-northwest-trending linear features that can be traced over a distance >250 km. These coarse-grained, ophitic dykes consist of euhedral plagioclase lathes (50%) in a matrix of subhedral to anhedral pyroxene, including zoned titanaugite and minor orthopyroxene. Quartz (5%), Fe–Ti oxides, and biotite are common, whereas chlorite and uralitic amphibole form patchy alteration zones.

Brown-weathering dykes of the 1998 Ma Minto swarm trend west-northwest to northwest and occur in bodies up to 60 m wide. They consist of coarse euhedral calcic

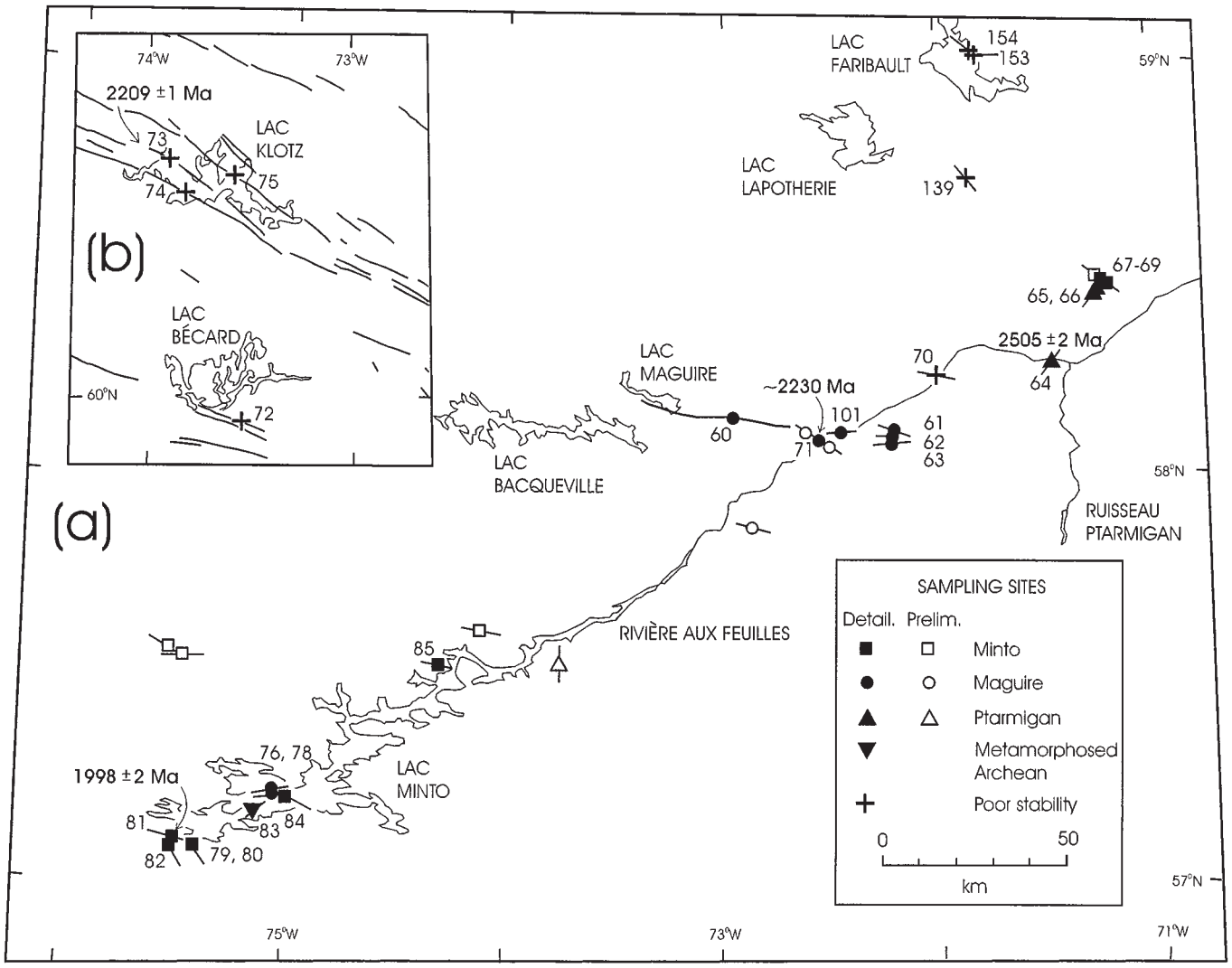
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Fig. 1. Diabase dyke swarms of the Minto block (see location on inset map) of the northeastern Superior Province and adjacent regions of the Canadian Shield. Dykes that can be assigned to specific swarms are shown as thick line segments, except for Minto dykes, which are displayed as dotted straight lines. Dykes that cannot presently be assigned to a particular swarm are shown as thin lines. Only U–Pb ages are indicated. B.I., Belcher Islands; C.S.B, Cape Smith Belt; L.T., Labrador Trough; R.G., Richmond Gulf. In the inset map boundaries of terranes in the Superior Province are shown as thin lines. Metasedimentary terranes are shaded. M.B., Minto block, N.C.T., North Caribou Terrane, N.Q.O., New Quebec Orogen; Phaner., Phanerozoic cover; U.O., Ungava Orogen.



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Fig. 2. Paleomagnetic and U–Pb dating sites in diabase dykes of the Minto block of Ungava Peninsula. (*a*) Rivière aux Feuilles and Lac Minto areas. (*b*) Lac Klotz area. Numbers are assigned only to detailed sites, from which three to seven samples were collected. Preliminary sites are those from which only a single sample was collected. Data from preliminary sites are not used in the statistical calculations. The area in which sites 65–69 are located is enlarged in Fig. 8.



plagioclase (60%) with subsidiary titanaugite (30%), Fe–Ti oxides (5%), and accessory apatite. Olivine occurs in proportions up to 30% and minor orthopyroxene is present in olivine-free dykes.

U–Pb geochronology

Analytical techniques

Samples for U–Pb geochronology were obtained from the coarse interiors of four of the thickest dykes, one in each of the Ptarmigan, Maguire, Minto, and Klotz swarms (Fig. 1, 2). Zircon and (or) baddeleyite were separated from 5–25 kg samples using conventional crushing, grinding, wet shaking table, heavy liquids, and magnetic separator methods. Individual fractions for analysis were selected on the basis of grain size and morphology. Analyses at the Geochronology Laboratory of the Geological Survey of Canada (Ottawa) used methods described by Mortensen (1993) and Buchan et al. (1993). All zircon fractions analysed were air abraded prior to dissolution. Baddeleyite grains were not abraded. In some samples, baddeleyite was too fine grained to abrade, or to determine whether grains were free of secondary zircon (Heaman and LeCheminant 1993). Analytical results are presented in Table 1 and shown with 2σ errors on conventional U–Pb concordia diagrams in Fig. 3.

Analytical results

Ptarmigan dyke

A small (~5 kg) sample of a >25 m thick Ptarmigan dyke (sample PBAS-90-114) was collected from paleomagnetic site 64 along Rivière aux Feuilles (Fig. 2*a*). Both zircon and baddeleyite were recovered from the sample. Zircon is present as elongate square to tabular prisms and skeletal grains, and baddeleyite occurs as thin, pale to medium reddish brown cleavage fragments.

Three fractions of strongly abraded zircon give nearly concordant (<1% discordant) analyses with overlapping

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Table 1. U–Pb analytical results.

Fraction[a]	Wt. (mg)	U (ppm)	Pb (ppm)[b]	$^{206}Pb/^{204}Pb$ (meas.)	Common Pb (pg)[c]	^{208}Pb	$^{206}Pb/^{238}U$[d]	$^{207}Pb/^{235}U$[d]	$^{207}Pb/^{206}Pb$[d]	$^{207}Pb/^{206}Pb$ age (Ma)[e]
Ptarmigan site 64 (PBAS-90-114)										
A,z,N2,>149	0.008	737	393	6 141	21	11.3	0.4719(0.08)	10.713(0.10)	0.4719(0.08)	2503.9(1.0)
B,z,N2,>149	0.003	384	219	1 119	31	17.2	0.4701(0.13)	10.662(0.14)	0.4701(0.13)	2502.6(2.2)
C,z,N2,>149	0.013	744	387	8 431	34	9.0	0.4725(0.08)	10.723(0.10)	0.4725(0.08)	2503.5(1.0)
E,b,N2,62–105,u	0.015	752	355	5 830	56	9.2	0.4670(0.09)	10.582(0.10)	0.4670(0.09)	2500.6(1.1)
F,b,N2,>74,u	0.023	610	287	6 253	64	2.1	0.4605(0.09)	10.369(0.10)	0.4605(0.09)	2490.3(1.0)
Maguire site 71 (MLB-91-015-b)										
B,b,N10,>74,u	0.038	353	135	8 087	38	4.8	0.3725(0.08)	7.0956(0.10)	0.1382(0.04)	2204.3(1.4)
C,b,N10,>74,u	0.020	313	123	9 392	16	4.6	0.3811(0.22)	7.2581(0.22)	0.1381(0.03)	2203.7(1.2)
D,b,N10,>74,u	0.017	309	122	7 123	18	3.5	0.3878(0.09)	7.4288(0.10)	0.1389(0.03)	2213.8(1.2)
E,z,N10,>105	0.006	1812	1601	7 760	23	69.4	0.2816(0.08)	4.3425(0.10)	0.1119(0.03)	1829.8(1.1)
F,z,N10,>105	0.008	1301	1043	6 313	32	61.3	0.3196(0.08)	5.4882(0.10)	0.1246(0.03)	2022.5(1.1)
G,z,N10,>105	0.014	1256	1057	8 586	42	61.1	0.3362(0.08)	5.9036(0.10)	0.1274(0.03)	2061.9(1.1)
Minto site 81 (MLB-91-023-a)										
B,b,N10,>105,u	0.030	69	23	455	98	3.0	0.3345(0.13)	5.5742(0.26)	0.1208(0.20)	1968.7(7.1)
C,b,N10,>105,u	0.020	50	17	505	42	5.4	0.3342(0.21)	5.5578(0.29)	0.1206(0.18)	1965.5(6.3)
D,b,N10,>105,u	0.040	63	21	976	54	4.3	0.3350(0.13)	5.5727(0.17)	0.1206(0.09)	1965.7(3.3)
E,z,N10,>74	0.013	715	349	8 447	24	28.6	0.3600(0.08)	6.0840(0.10)	0.1226(0.03)	1994.2(1.1)
G,z,N10,>74	0.022	436	213	12 840	17	28.5	0.3607(0.08)	6.1036(0.10)	0.1227(0.03)	1996.2(1.1)
H,z,N10,>74	0.019	430	213	1 042	182	29.6	0.3608(0.09)	6.1008(0.15)	0.1226(0.10)	1995.1(3.4)
Klotz site 73 (MLB-91-017)										
A,b,N2,>160,u	0.018	92	37	1 517	27	4.8	0.3934(0.13)	7.4880(0.15)	0.1380(0.06)	2202.7(1.9)
B,b,N2,>160,u	0.027	156	63	1 281	82	4.4	0.3977(0.10)	7.5817(0.13)	0.1383(0.07)	2205.5(2.4)
C,b,N2,>160,u	0.025	127	51	1 628	47	4.4	0.3887(0.25)	7.3791(0.27)	0.1377(0.07)	2198.4(2.5)
D,b,N2,>160,u	0.035	149	63	1 420	87	12.9	0.3773(0.11)	7.1025(0.14)	0.1365(0.07)	2183.5(2.4)
E,z,N5,>160	0.156	290	147	17 130	7	21.0	0.4079(0.08)	7.7941(0.10)	0.1386(0.03)	2209.6(0.9)
F,z,N5,>160	0.120	315	161	10 100	10	21.6	0.4086(0.10)	7.8060(0.11)	0.1385(0.03)	2209.1(0.9)

[a]b, baddeleyite; z, zircon; N2, N5, N10, nonmagnetic at given degrees side slope on Frantz isodynamic magnetic separator; grain size given in microns; u, unabraded.
[b]Radiogenic Pb; corrected for blank, initial common Pb, and spike.
[c]Corrected for spike and fractionation.
[d]Corrected for blank Pb and U, and common Pb; errors are in percent at 1σ level.
[e]Errors are in Ma at a 2σ level.

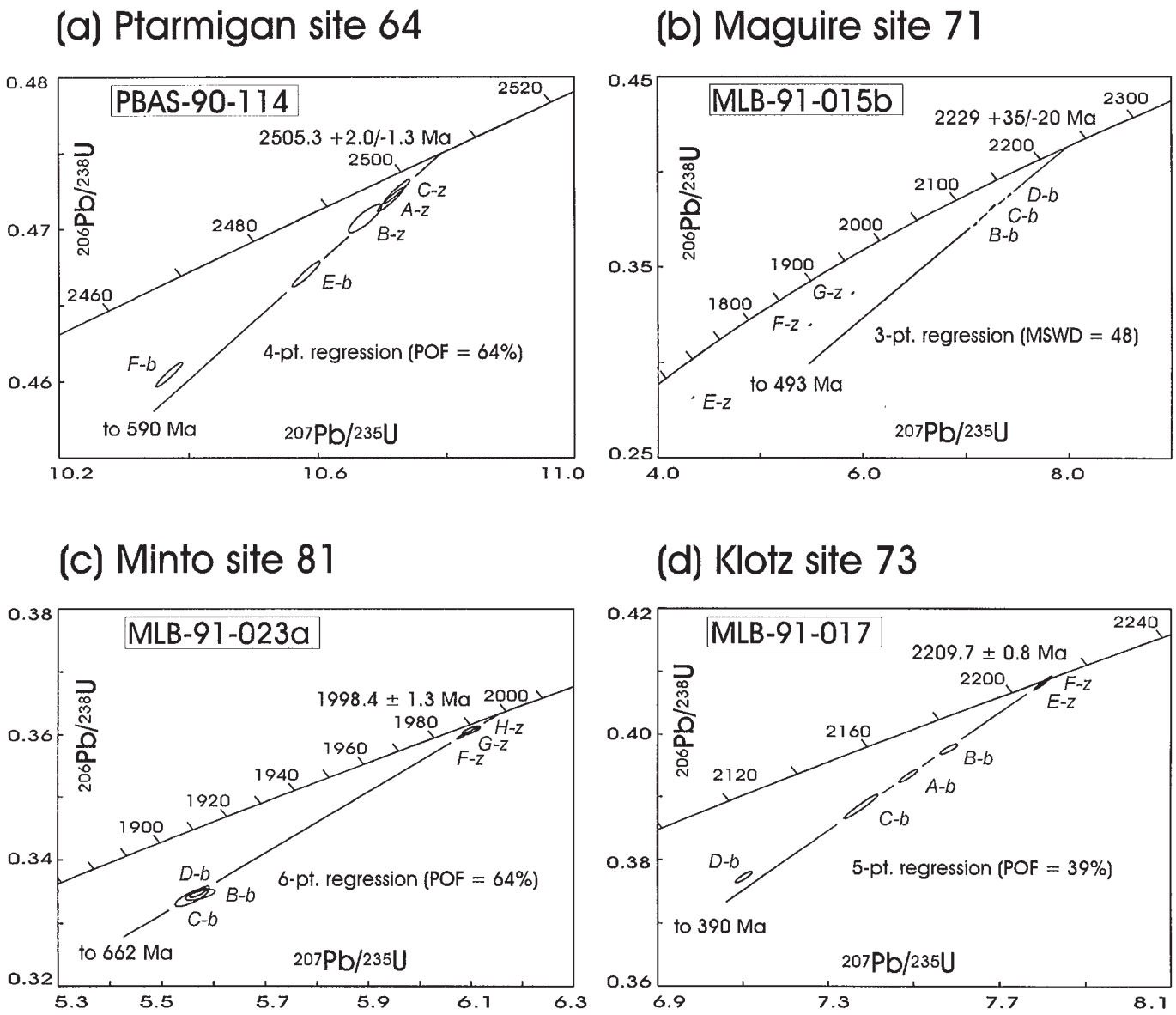
$^{207}Pb/^{206}Pb$ ages of 2503–2504 Ma. Two fractions of baddeleyite were also analysed; fraction E is 1.4% discordant and gives a $^{207}Pb/^{206}Pb$ age similar to those of the zircon fractions. The surfaces of the baddeleyite grains in fraction E were very smooth and vitreous and showed no evidence of alteration. However, some of the grains in the second baddeleyite fraction (F) had slightly frosted surfaces, which may indicate a very thin layer of secondary zircon replacing the baddeleyite. This analysis was 2.3% discordant and gave a slightly younger $^{207}Pb/^{206}Pb$ age. A Davis-type regression (Davis 1982) through the three zircon fractions and baddeleyite fraction E gives intercept ages of $2505.3^{+2.0}_{-1.3}$ Ma and 590 Ma, with a 64% probability of fit (Fig. 3*a*). The upper intercept age is interpreted as the crystallization age of the sample, and the lower intercept indicates that the minor Pb loss from zircon and baddeleyite was of latest Proterozoic age. Because of the possibility that baddeleyite fraction F may have included a minor component of secondary zircon, this analysis was excluded from the regression.

Maguire dyke

A 45 m thick Maguire dyke (sample MLB-91-015b) was sampled at paleomagnetic site 71 on Rivière aux Feuilles (Fig. 2*a*). Both baddeleyite and zircon were recovered. Analyses of three fractions of unabraded baddeleyite (Fig. 3*b*) are moderately discordant (5–9%). The baddeleyite grains appeared to be unaltered; therefore, it is unclear whether the discordance is due to the presence of secondary zircon or postcrystallization lead loss. The baddeleyite analyses define a roughly linear array. However, a Davis-type regression is inappropriate for analyses with this degree of discordance. A modified York II model regression (Parrish et al. 1987) yields intercept ages of 2229^{+35}_{-20} Ma and 493 Ma, with a mean square weighted deviation of 48. Zircon in this rock forms stubby square prisms with abundant internal fractures and both clear and opaque inclusions. Analyses of three fractions of abraded zircon are strongly discordant (10–14%). The three zircon analyses also form a roughly linear array, with highly imprecise intercept ages of ~2290 and ~1270 Ma. The upper concordia intercept defined by the

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Fig. 3. U–Pb dating of diabase dykes of the Minto block. (*a*) Ptarmigan, (*b*) Maguire, (*c*) Minto, and (*d*) Klotz dykes. b, baddeleyite fraction; z, zircon fraction; errors are given at a 2σ level.



baddeleyite analyses, although imprecise, is considered to be the best estimate of the crystallization age of the dyke. The similar upper intercept age determined from the zircon analyses suggests that the zircon and baddeleyite crystallized at the same time, but that zircon experienced a Mesoproterozoic Pb-loss event that apparently did not affect baddeleyite.

Minto dyke

Zircon and baddeleyite were recovered from a 60 m thick Minto dyke (sample MLB-91-023a) on Lac Minto at paleomagnetic site 81 (Fig. 2*a*). Three fractions (B, C, and D) of relatively coarse, clear, high-quality baddeleyite yield moderately discordant (6%) analyses, whereas three fractions (F, G, and H) of strongly abraded zircons yield nearly concordant (0.5–0.7% discordant) analyses with $^{207}Pb/^{206}Pb$ ages from 1996 to 1994 Ma (Fig. 3*c*). A Davis regression through the zircon and baddeleyite analyses gives intercepts of 1998.4 ± 1.3 Ma and 662 Ma. Because of the clustering of the data, this is essentially a two-point regression; however, the $^{207}Pb/^{206}Pb$ ages of the zircon fractions are only ~3 Ma younger than the calculated upper intercept, supporting our interpretation that the upper intercept gives the best estimate for the crystallization age of the dyke.

Klotz dyke

A sample was collected from an 80 m thick Klotz dyke (sample MLB-91-017) at paleomagnetic site 73 on Lac Klotz (Fig. 2*b*). Relatively coarse-grained, high-quality baddeleyite was recovered from the sample, together with elongate and prismatic to skeletal, clear zircon grains. Four fractions of baddeleyite (fractions A–D) are moderately discordant (2.5–6.4%) with $^{207}Pb/^{206}Pb$ ages from 2205 to 2183 Ma (Fig. 3*d*). Two fractions (E and F) of abraded zircon are concordant, with identical ages. The weighted average of the $^{207}Pb/^{206}Pb$ ages for the two fractions is 2209.4 ±

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Table 2. Summary of characteristic remanences for mafic dyke sites of central Ungava.

Site	Swarm	N (n_A,n_T)	Lat. (°N)	Long. (°W)	Width (m)	Trend (°)	D (°)	I (°)	D′ (°)	I′ (°)	k	α_{95} (°)
60	Maguire	5 (5,0)	58.15	72.95	150	290	19.6	–35.7	18.5	–36.3	36	13.0
61	Maguire	5 (4,0)	58.12	72.20	13	290	28.3	–47.8	26.3	–48.5	460	9/21
62	Maguire	5 (4,0)	58.10	72.21	5	275	20.9	–32.8	19.1	–33.5	105	9.0
63	Maguire	5 (5,0)	58.08	72.22	>50	260	27.3	–42.6	25.4	–43.3	217	5.2
64[a]	Ptarmigan	5 (5,0)	58.28	71.46	>25	220	50.0	49.5	48.3	48.1	170	5.9
65	Ptarmigan	5 (5,0)	58.44	71.26	45	220	53.8	35.9	51.7	33.7	50	10.9
66	Ptarmigan	5 (5,0)	58.45	71.25	45	220	47.5	44.5	45.5	42.7	61	9.9
67	Minto	5 (5,0)	58.46	71.19	>13	310	293.1	49.6	291.4	50.9	354	4.1
68	Minto	5 (5,0)	58.46	71.21	>20	310	289.4	46.9	287.6	48.3	235	5.0
69	Minto	5 (4,0)	58.47	71.23	~30	~310	297.3	44.6	295.5	45.9	64	9.6
70[b]	Maguire?	6 (0,0)	58.25	72.01	22	280	—	—	—	—	—	—
71[a]	Maguire	5 (4,0)	58.09	72.54	45	310	5.3	–37.9	3.9	–38.1	141	7.8
72	Klotz?	5 (0,0)	59.95	73.54	>75	285	—	—	—	—	—	—
73[a]	Klotz	3 (0,0)	60.58	73.89	80	305	—	—	—	—	—	—
74	Klotz	5 (0,0)	60.50	73.83	80	300	—	—	—	—	—	—
75	Klotz	5 (0,0)	60.54	73.58	90	305	—	—	—	—	—	—
76	Maguire	5 (3,0)	57.22	75.03	2	270	36.3	–54.2	37.2	–53.2	44	18.7
78	Maguire	5 (4,1)	57.22	75.03	15	260	21.4	–27.5	22.4	–25.8	53	10.6
79[c]	Minto	7 (2,1)	57.09	75.41	2[c]	~280	141.4	–25.8	142.3	–26.2	71	14.7
80	Minto	4 (2,2)	57.09	75.41	1	325	141.2	–31.5	142.0	–31.9	28	17.6
81[a]	Minto	7 (3,1)	57.10	75.49	60	~105	128.9	–27.3	129.8	–27.2	71	11.0
82	Minto	4 (3,0)	57.09	75.50	0.2	330	136.5	–60.3	136.7	–60.4	94	12.8
83	Deformed Archean	4 (4,0)	57.17	75.12	—	230	226.1	–20.7	227.0	–22.6	45	13.9
84	Minto	6 (1,4)	57.21	74.96	~15	295	135.4	–20.7	136.0	–21.0	242	5.9
85	Minto	5 (1,4)	57.53	74.30	17	280	125.4	–27.4	125.6	–27.6	103	9.1
101	Maguire	4 (2,2)	58.11	72.44	12	265	19.6	–25.9	18.2	–26.5	18	22.2
139	?	3 (0,0)	58.73	71.86	0.5	320	—	—	—	—	—	—
153	?	4 (0,0)	59.02	71.81	>1.5	270	—	—	—	—	—	—
154	?	3 (0,0)	59.03	71.82	35	310	—	—	—	—	—	—

Notes: N is number of samples collected; n_A is number of AF-demagnetized samples utilized in mean calculation based on analysis of Kirschvink (1980), with the exception of site 61, where the mean calculation is based on remagnetization circle analysis (Bailey and Halls 1984); n_T is number of thermally demagnetized samples utilized in mean calculation. Width and trend of dyke are measured at sampling site. D and I are mean declination and inclination of characteristic remanence; D′ and I′ are mean declination and inclination recalculated for common reference locality of 58.0°N, 74.0°W. k is precision parameter. α_{95} is radius of circle of 95% confidence, with the exception of site 61, for which confidence parameters associated with remagnetization circle analysis (Bailey and Halls 1984) are given.

[a]Site dated by the U–Pb technique in this study.

[b]At this site some samples show poorly defined Maguire directions, but were too poorly constrained to allow a calculation of a mean direction.

[c]Site consists of samples from three closely spaced plagioclase porphyritic dykes (average width of 2 m) that crosscut the nonporphyritic dyke of site 80.

0.6 Ma, which we consider to be the best estimate for the crystallization age of the rock. A Davis regression through baddeleyite fractions A, B, and C and the two zircon fractions gives intercept ages of 2209.7 ± 0.8 and 390 Ma with a 39% probability of fit (Fig. 3*d*). Baddeleyite fraction D may have been slightly altered during a Neoproterozoic or early Paleozoic alteration event that did not affect baddeleyite in the other three fractions.

Paleomagnetism

Sampling and laboratory techniques

Preliminary paleomagnetic work based on single oriented blocks from each of 21 sites near Rivière aux Feuilles and Lac Minto identified three distinct directions of magnetization. To better define these paleomagnetic directions, detailed paleomagnetic sampling (three to seven samples per site) was conducted at 24 sites near Rivière aux Feuilles and Lac Minto (Fig. 2*a*), and at four sites in the set of prominent northwest-trending Klotz dykes about 250 km north of Rivière aux Feuilles at Lac Klotz and Lac Bécard (Fig. 2*b*). A single deformed Archean dyke was sampled for paleomagnetism at Lac Minto to determine whether such units are suitable for Neoarchean paleomagnetic study.

In the detailed paleomagnetic study, samples were oriented with sun and magnetic compasses. Those in the preliminary work were oriented with only a magnetic compass. Each sample was drilled in the laboratory to produce two or more cylindrical specimens with diameters of 2.5 cm and lengths of 2.2 cm. Specimens were measured using a Schonstedt spinner magnetometer and subjected to stepwise alternating fields (AF's), thermal step heating, or a combination of AF demagnetization followed by thermal cleaning. All measurements were carried out in a shielded room with residual magnetic field <3000 nT. Paleomagnetic directional data for individual specimens were analysed using

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Table 3. Mean paleomagnetic results for Ungava dyke swarms.

Dyke swarm	No. of sites	D′ (°)	I′ (°)	*k*	α_{95} (°)	Plat. (°N)	Plong. (°E)	dm (°)	dp (°)
Ptarmigan[a]	3	48.6	41.5	112	11.7	–41.6	219.5	14.3	8.7
Maguire	8	20.7	–38.4	46	8.3	–8.5	266.6	9.8	5.8
Minto[b]	9	128.5	–38.3	23	11.0	38.2	173.8	13.1	7.7

Notes: D′ and I′ are declination and inclination corrected to a common reference location at 58.0°N, 74.0°W. *k* is precision parameter. α_{95} is radius of circle of 95% confidence. Plat. and Plong. are coordinates of the mean pole. dm and dp are semiaxes of the oval of 95% confidence about the mean pole.

[a]The Ptarmigan calculation yields a virtual geomagnetic pole because it is based on only three sampling sites, insufficient to average out secular variation.

[b]Directions for three Minto sampling sites (67, 68, and 69) with down west-northwest directions have been inverted in order to calculate the overall Minto mean direction.

Kirschvink (1980) analysis and remagnetization circle analysis (Bailey and Halls 1984).

A paleomagnetic baked contact test was conducted at northwest-trending Maguire dyke site 71. Oriented samples were collected from the dyke and from host rocks both adjacent to and far from the dyke. At another location three paleomagnetic sites (67, 68, and 69) were collected in a northwest-trending Minto dyke and two sites (65 and 66) in an older Ptarmigan dyke that trends northeast toward it. A rigorous baked contact test was not possible because the older dyke is not exposed adjacent to the younger dyke, and, hence, baked samples of the older dyke could not be collected.

Results

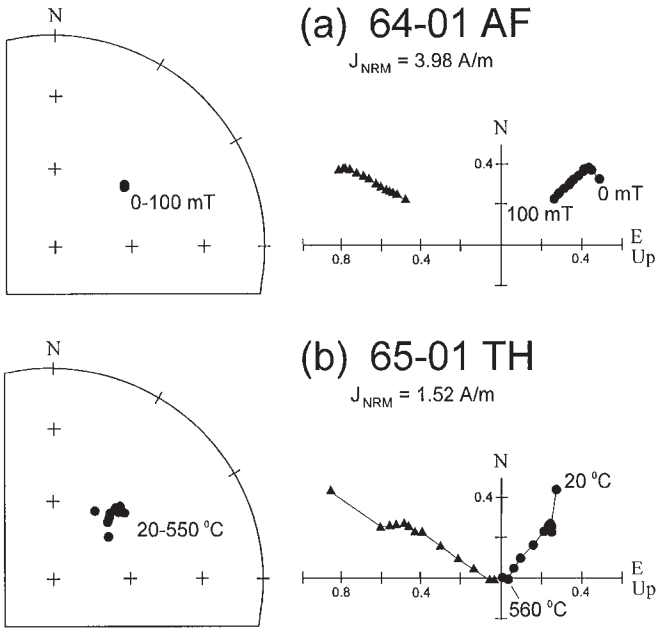
The detailed paleomagnetic study confirmed the presence of three distinct magnetization directions in dykes collected from the Rivière aux Feuilles – Lac Minto area, whereas the results from the dykes near Lac Klotz and Lac Bécard were more difficult to interpret. The data are summarized in Tables 2 and 3.

Ptarmigan dykes

Detailed paleomagnetic results for the 2505 Ma north- to northeast-trending Ptarmigan swarm were obtained at site 64 on Rivière aux Feuilles near the junction of Ruisseau Ptarmigan and at sites 65 and 66 about 10 km to the northeast (Fig. 2*a*). Sites 65 and 66 are located along the same dyke. The paleomagnetic direction at all three sites is very stable and directed down to the northeast, with little evidence of a significant viscous component (Fig. 4). Thermal demagnetization experiments indicate that the stable magnetization is carried by minerals with blocking temperatures of 500–580°C (Fig. 4*b*), consistent with low-titanium titanomagnetite as the remanence carrier. The single sample collected at a preliminary site in a north-trending dyke about 160 km to the southwest (Fig. 2*a*) yields a somewhat similar paleomagnetic direction, suggesting that this dyke may also belong to the Ptarmigan swarm. Only a few other dykes of northeast to north trend have been mapped in the Minto block (Fig. 1).

The Ptarmigan magnetization has not been demonstrated to be primary with a paleomagnetic field test. However, as discussed below, paleomagnetic evidence indicates that it predates the emplacement of the Minto dyke swarm at 1998 Ma.

Fig. 4. Typical examples of paleomagnetic data for Ptarmigan dyke specimens upon (*a*) AF demagnetization and (*b*) thermal demagnetization. Closed (open) circles on equal-area projections indicate downward (upward) directions. Circles (triangles) on vector diagrams represent projection onto the horizontal (vertical) plane. J_{NRM} is the intensity of natural remanent magnetization.



Maguire dykes

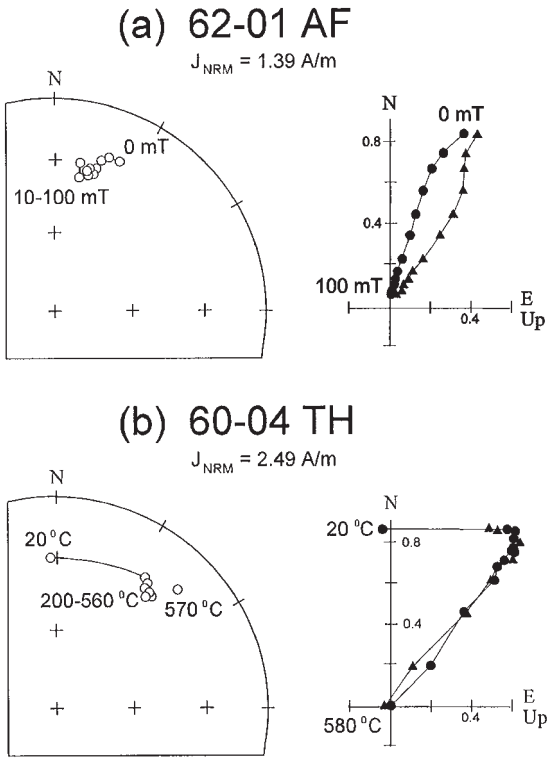
West-northwest-trending Maguire dykes, dated at ca. 2230 Ma, carry a stable remanence directed up to the north-northeast (Fig. 5). They occur in abundance in the region between Lac Maguire and Rivière aux Feuilles (Fig. 2*a*), but are also present to the west in the Lac Minto area. Nine Maguire sites have been studied in detail, with eight yielding results suitable for statistical analysis. Three preliminary sites (Fig. 2*a*) yield similar remanence directions.

Stable Maguire directions are usually obtained after removal of a viscous component (Fig. 5). Thermal demagnetization experiments yield blocking temperatures for the stable component of 500–580°C (Fig. 5*b*), indicating that low-titanium titanomagnetite carries the stable remanence.

A positive baked contact test at site 71 (Fig. 6) indicates that the stable Maguire remanence is likely primary. Four of the five samples collected from the dyke are quite stable on AF demagnetization, whereas in host samples stable direc-

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Fig. 5. Typical examples of paleomagnetic data for Maguire dyke specimens upon (*a*) AF demagnetization and (*b*) thermal demagnetization. Symbols and explanations as in Fig. 4.

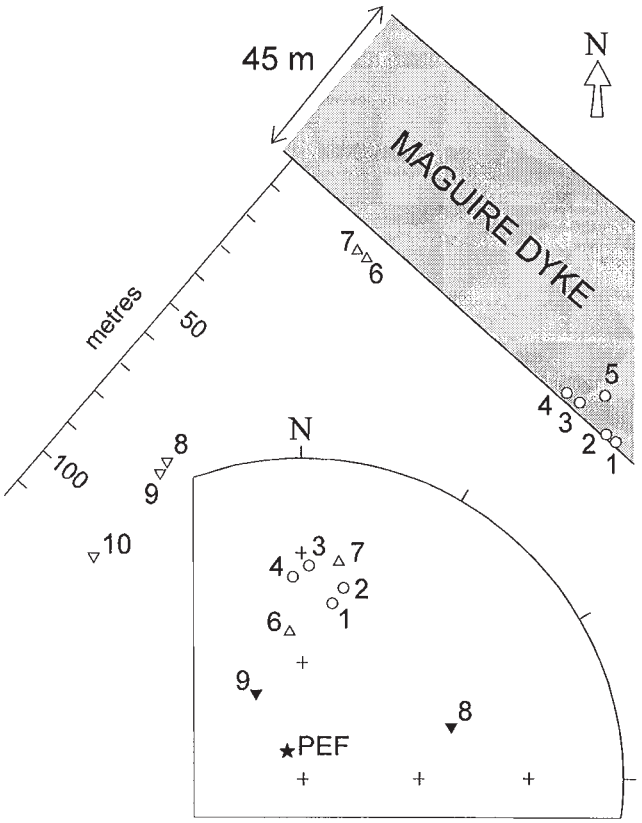


tions are best isolated by thermal demagnetization. Samples 6 and 7, collected in mafic Archean rocks 7 m from the 45 m wide dyke, yield remanence directions similar to that of the dyke, indicating that they were baked at the time of dyke emplacement. On the other hand, samples 8 and 9, at 80 and 83 m respectively, carry stable but discordant magnetizations with down directions. Blocking temperatures of samples 8 and 9 are quite high (530–570°C), suggesting that the remanence of these samples predates that of the dyke rather than being of viscous origin acquired along the present Earth's magnetic field. Sample 10, at 110 m, was difficult to measure because of its weak natural remanent magnetization (NRM). Nevertheless, its NRM direction is similar to that of sample 9.

Minto dykes

West-northwest- to northwest-trending Minto dykes, dated at 1998 ± 2 Ma, carry a dual polarity remanence directed up to the southeast (reversed) at six sampling sites (Figs. 7*a*, 7*b*) and down to the west-northwest (normal) at three sites from a single dyke (Figs. 7*c*, 7*d*). Similar directions were observed at four other preliminary sites (Fig. 2*a*). Minto dykes are most abundant in the Lac Minto area, but also occur to the east near Rivière aux Feuilles (Fig. 2*a*). Minto remanence directions are typically isolated after elimination of a viscous remanence (Fig. 7). Thermal demagnetization of both normal and reversely magnetized samples indicates that the remanence is carried by low-titanium titanomagnetite.

Fig. 6. Baked contact test at Maguire dyke site 71. Circles indicate samples located in the dyke, upright triangles indicate those in the baked host Archean mafic rocks, and inverted triangles indicate those in the unbaked host Archean mafic rocks. Stable directions based on Kirschvink (1980) analysis of AF demagnetization data for dyke samples, and thermal demagnetization data for host samples are shown on the equal area plot, with closed (open) symbols indicating downward (upward) directions. The star labelled PEF represents the direction of the present Earth's magnetic field.



A baked contact test was not carried out for the Minto dykes. However, a dyke carrying the Minto direction at sites 67, 68, and 69 and a dyke carrying the Ptarmigan direction at sites 65 and 66 (Fig. 8) were sampled in close proximity to one another. The younger Minto dyke strikes perpendicular to and cuts across the trend of the Ptarmigan dyke, but the Ptarmigan dyke is not exposed adjacent to the Minto dyke, excluding the possibility of a baked contact test. Although the magnetizations of the two dykes have similar blocking temperatures, their directions are quite distinct (Table 2). They cannot be explained as an overprint after emplacement of the younger Minto dyke, because the direction of such an overprint would be the same in both dykes. Therefore, the Minto dyke remanence is likely primary, and the Ptarmigan dyke remanence must be older than the emplacement age of the Minto dyke.

Stable remanence directions obtained at detailed sites in Ptarmigan, Maguire, and Minto dykes are summarized in Fig. 9. Because of the dramatic difference among the three directional groupings, assigning dykes to these swarms is straightforward based on their paleomagnetic directional characteristics.

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Fig. 7. Typical examples of paleomagnetic data for Minto dyke specimens. (*a*) A reversely magnetized specimen on AF demagnetization, (*b*) a reversely magnetized specimen on thermal demagnetization, (*c*) a normally magnetized specimen on AF demagnetization, and (*d*) a normally magnetized specimen on thermal demagnetization. Symbols and explanations as in Fig. 4.

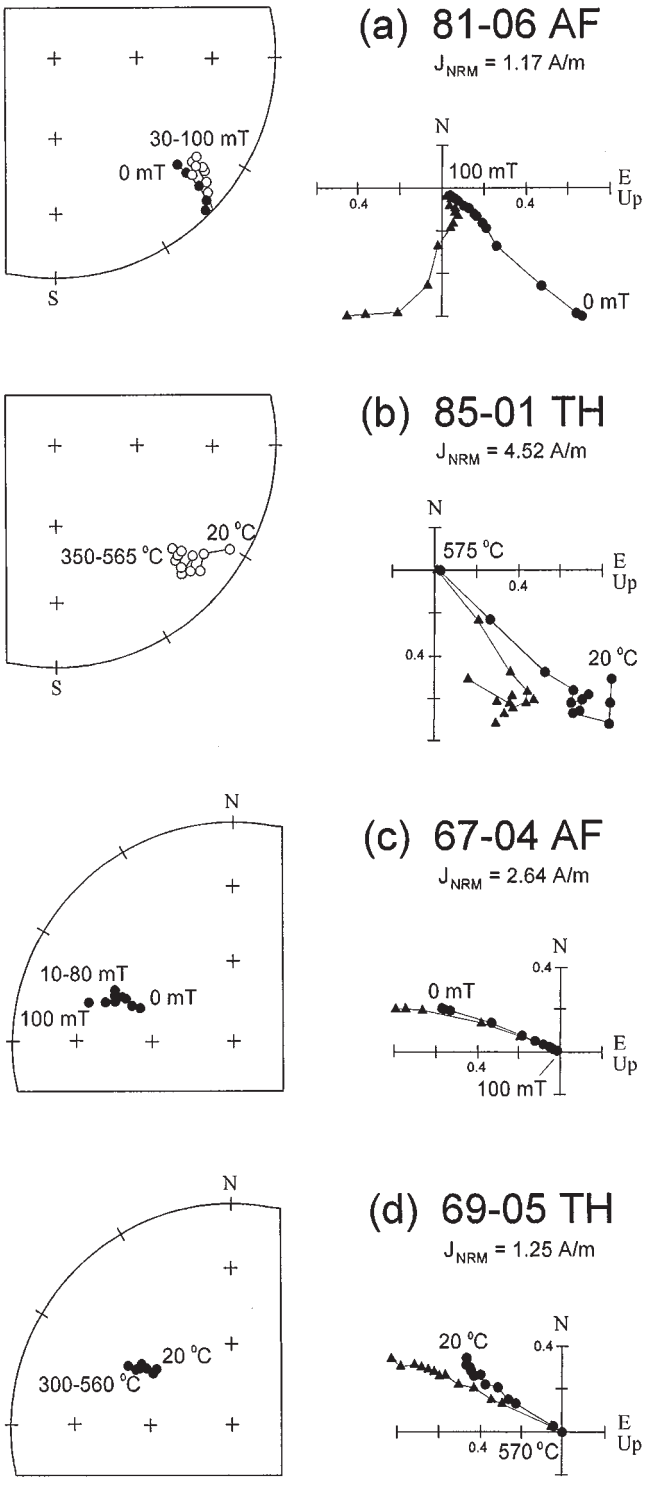
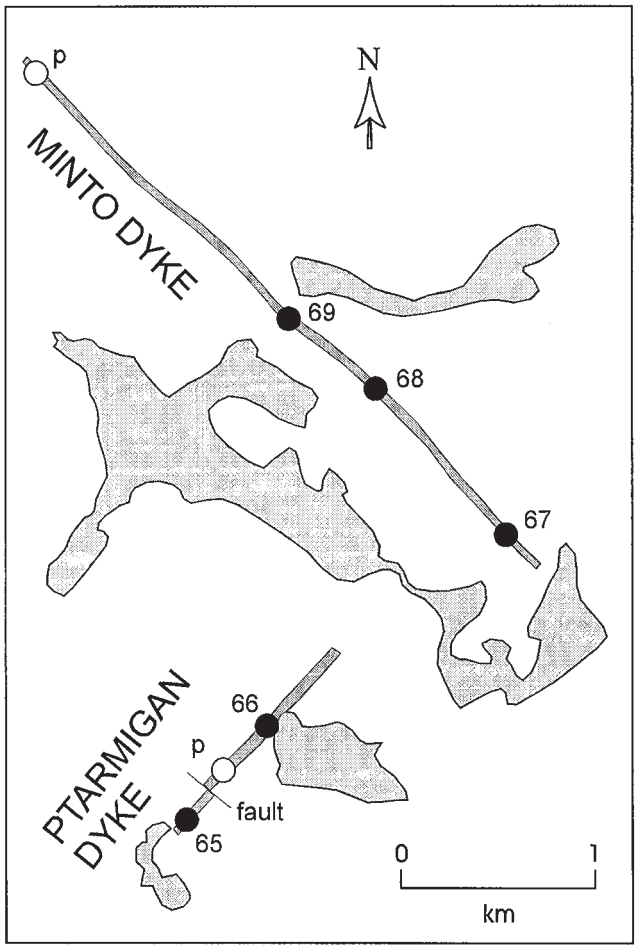


Fig. 8. Paleomagnetic sampling sites in a Minto dyke and a nearby Ptarmigan dyke. p, preliminary sampling site (see definition in Fig. 2); lakes and ponds are shaded.

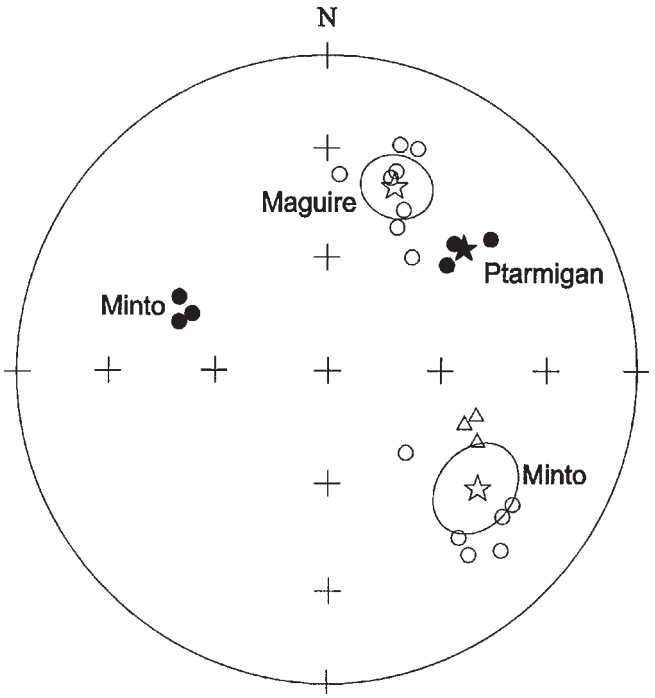


Klotz dykes

None of the four paleomagnetic sampling sites in the 2209 Ma Klotz swarm at Lac Klotz and Lac Bécard (Fig. 2*b*) yields a stable, consistent remanence direction. Site 75 is unstably magnetized. At sites 72, 73, and 74 (Fig. 10), directions from individual specimens fall along a rough great circle path between a down southwest direction and a shallow up north-northeast direction (Fig. 10*a*). Many individual specimens from these sites show directional swings along this path on AF or thermal demagnetization (Fig. 10*b*). The northeast end point of the path may indicate that these sites carry a primary remanence direction like that of similar-aged Maguire dykes. If so, the smearing of directions towards a down southwest direction may reflect superposition of a stable secondary component. The secondary component could be a ca. 1.8 Ga "Hudsonian" overprint, similar to that observed in a number of units in the Circum-Ungava fold belt (e.g., Schmidt 1980; Buchan and Baragar 1985) or a reversed Maguire direction. A partial Hudsonian overprint on Klotz dykes, but not on Maguire dykes, could be due to the proximity of Klotz dykes to the Cape Smith Belt of the Circum-Ungava fold belt or to differences in magnetic characteristics. No definite conclusion can be

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Fig. 9. Summary of stable paleomagnetic directions from Ptarmigan, Maguire, and Minto dyke sites (circles). Triangles represent the antipoles of normally magnetized Minto sites. Stars indicate mean directions for each dyke swarm. Circles of 95% confidence about the means are also shown. Other symbols as in Fig. 4.



drawn from the present results. It should be noted that the present Earth's magnetic field direction also falls near the great circle (Fig. 10). However, a viscous component is an unlikely explanation for the southwest component because of its high coercivities (up to 100 mT) and high blocking temperatures (up to 570°C).

Deformed Archean dyke

The deformed Archean dyke, which was sampled at site 83, carries a stable remanence with a shallow southwest direction (Table 2). Its remanence was likely set during cooling from ca. 2.7 Ga granulite-facies metamorphism in this part of the Minto block.

Discussion

The present study has identified diabase dyke swarms of three distinct ages in the Minto block. Olivine diabase Ptarmigan dykes, dated at 2505 ± 2 Ma, form a sparse swarm trending north to northeast. Intrusions and volcanics that are coeval or slightly younger than the Ptarmigan dykes are common in other parts of the Canadian and Baltic shields and may indicate an intense and widespread episode of magmatism in the latest Archean and earliest Proterozoic (Heaman 1997). They include the 2473–2446 Ma (U–Pb baddeleyite; Heaman 1997) Matachewan dyke swarm of the southern Superior Province and related 2490–2450 Ma volcanics and intrusions (summarized by Prevec et al. 1995), the ca. 2470 Ma (Heaman 1994) Mistassini swarm of the southern Superior Province, the ca. 2450 Ma (Heaman 1994) Kaminak swarm of the Churchill Province, and major episodes of mafic–ultramafic magmatism at ca. 2505–2433 Ma in the northeastern Baltic Shield (summarized by Smolkin 1997).

The quartz diabase Maguire and Klotz dykes have similar ages, 2229^{+35}_{-20} and 2209 ± 1 Ma respectively, and may represent subswarms of the same dyke swarm. The Klotz dykes have a well-defined west-northwest trend, whereas the short Maguire dyke segments that have been identified to date trend west to northwest, with west to west-northwest orientations dominating. Klotz–Maguire-aged intrusions are common in the Superior and Nain provinces. The Senneterre dyke swarm of the southern Superior Province has been dated at ca. 2216 Ma (Buchan et al. 1993; J.K. Mortensen unpublished data referred to in Buchan et al. 1996), the Nipissing sills that intrude sedimentary rocks of the Huronian Supergroup of the Southern Province at 2217–2210 Ma (Andrews et al. 1986; Noble and Lightfoot 1992), and the Kikkertavak dyke swarm of the Nain Province at 2235 Ma (Cadman et al. 1993).

Minto dykes are dated at 1998 ± 2 Ma. They trend from west-northwest to northwest and only olivine-bearing members are distinct in the field from similarly trending Maguire and Klotz dykes. The age determined herein for the Minto dykes is indistinguishable from the U–Pb zircon age of 1998 ± 2 Ma reported for the Watts Group of the Purtuniq ophiolite in the Cape Smith Belt (Parrish 1989) to the north of the study area.

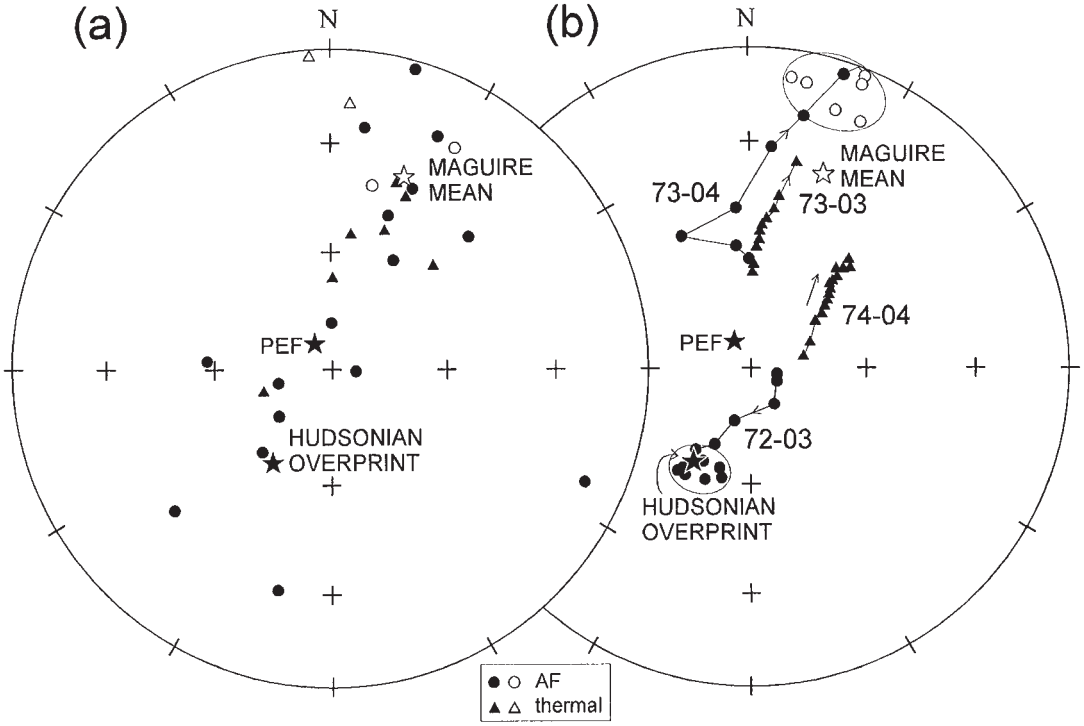
At present it is difficult to assign unsampled dykes of the study area (Fig. 1) to the dated swarms. Indeed, it is likely that some of the unsampled dykes belong to other swarms identified in the surrounding regions. These include the prominent north-northwest- trending Payne River swarm immediately west of Ungava Bay (Fahrig et al. 1986), a few dykes in and north of the Cape Smith Belt that are thought (Fahrig and West 1986) to belong to the large 723 Ma Franklin swarm (Heaman et al. 1992), and the undated northwest- trending Inukjuak dykes that outcrop along the east coast of Hudson Bay (Legault et al. 1994).

As illustrated in Fig. 9, distinct paleomagnetic directions have been obtained from Ptarmigan, Maguire, and Minto dykes. For several reasons we interpret the Maguire and Minto remanences to be primary and the Ptarmigan remanence to be primary or to have been acquired not long after emplacement of the swarm. First, dating at sites with the different remanence directions has yielded distinct ages. Second, the baked contact test for the Maguire dyke (Fig. 6) and the comparison of directions from Minto and Ptarmigan dykes only a few hundred metres apart (Fig. 8) suggest, while not entirely conclusively, that the Maguire and Minto remanences are primary and that the Ptarmigan remanence predates emplacement of Minto dykes. Third, Maguire and Ptarmigan paleopoles are similar to those of distant units of similar age in the Superior and Southern provinces as described below. Finally, the Ptarmigan dykes intruded at 2505 Ma when regional cooling temperatures had reached about 300°C as indicated by ca. 2.5 Ga K–Ar mica ages, so that the Ptarmigan remanence with blocking temperatures of 500–580°C was likely acquired at that time.

The deformed Archean dyke at site 83 carries a remanence that is reversed to, but only about 20° different in

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Fig. 10. Paleomagnetic data for Klotz dykes. (*a*) Scatter of specimen directions between mean direction for Maguire dykes and expected Hudsonian overprint direction based on that observed in the Circum-Ungava fold belt. (*b*) Directional swings from individual specimens. Symbols and explanations as in Fig. 4.



direction (Table 2) from, that of the Ptarmigan dykes. Therefore, although the Archean dyke was undoubtedly emplaced before the ca. 2.7 Ga metamorphism and magnetized during regional cooling, the similarity of its remanence direction with that of the Ptarmigan dykes suggests that it was magnetized when the Superior Province had a latitude and orientation similar to those when the Ptarmigan dykes were magnetized.

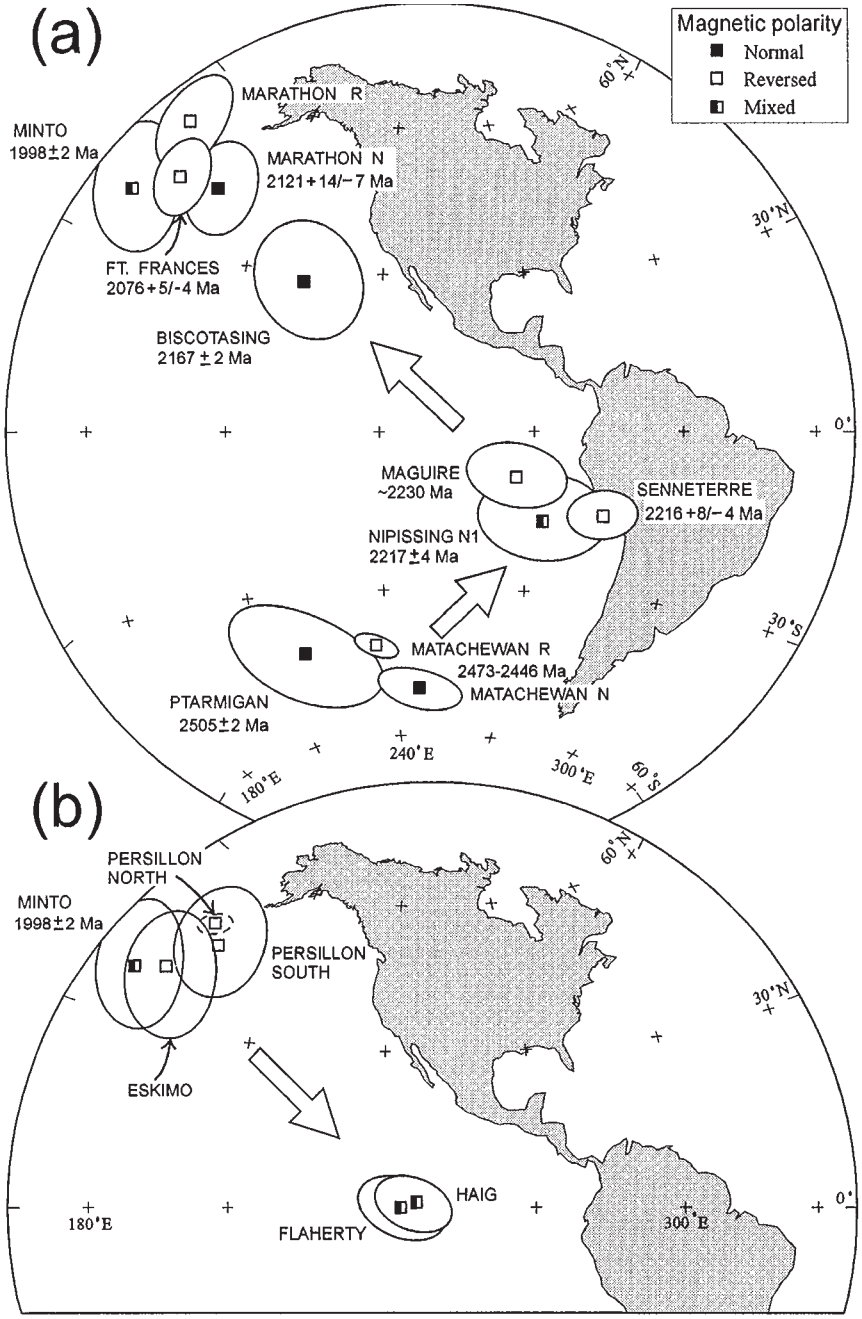
The ca. 2230 Ma Maguire dykes have a paleomagnetic pole, based on eight sampling sites, of 8.5°S, 266.6°E (dm = 9.8°, dp = 5.8°), similar to primary (e.g., Buchan 1991) paleopoles reported for ca. 2216 Ma Senneterre dykes of the southern Superior Province and ca. 2217 Ma N1 Nipissing sills of the Southern Province (Fig. 11*a*). The similarity of the ca. 2.22 Ga paleopoles from the two areas indicates that no measurable relative rotation or latitudinal movement has taken place between the Minto block and the southern Superior Province since emplacement of these units. Therefore, the orientation of the structural trends of the Minto block and southern Superior Province, which are nearly at right angles to one another (insert of Fig. 1), predates 2.22 Ma. This supports the model of Percival et al. (1994) in which the difference in structural trends is considered to be a primary feature.

Ptarmigan dykes have a virtual geomagnetic pole of 41.6°S, 219.5°E (dm = 14.3°, dp = 8.7°) that is in general agreement with primary paleopoles from the slightly younger (2473–2446 Ma) Matachewan dykes of the southern Superior Province (Fig. 11*a*). However, the Ptarmigan pole, based on only three sampling sites from two dykes, does not average out secular variation, so a rigorous comparison of the Parmigan and Matachewan poles cannot be made at this time. Nevertheless, the general agreement between the ca. 2.50–2.45 Ga poles from the two areas suggests that the southern Superior Province and Minto block have maintained approximately their present relative latitude and azimuthal orientation since that time, and the difference in structural trends between the two areas predates the end of the Archean.

The 1998 Ma Minto dykes have a paleomagnetic pole, based on six reversely magnetized and three normally magnetized sampling sites, of 38.2°N, 173.8°E (dm = 13.1°, dp = 7.7°). No paleomagnetic poles that are demonstrably coeval with the 1998 Ma Minto swarm are yet available from units cutting the southern Superior Province. However, the paleopole for the Minto dykes (Fig. 11*a*) is similar to primary paleopoles obtained from the significantly older 2121^{+14}_{-7} Ma (U–Pb; Buchan et al. 1996) Marathon and 2076^{+5}_{-4} Ma (U–Pb; Buchan et al. 1996; Wirth et al. 1995) Fort Frances dykes (Halls 1986) of the southern Superior Province (Fig. 11*a*). In addition, the Minto paleopole is similar to prefold paleopoles reported from the Persillon South volcanics of Richmond Gulf that are thought to postdate 2025 ± 25 Ma (Chandler and Parrish 1989) and the Eskimo volcanics of the Belcher Islands that have been correlated with the Persillon Formation on the basis of paleomagnetism (Schwarz and Fujiwara 1981) and geochemistry (Legault et al. 1994) (Fig. 11*b*). Inukjuak dykes (Fig. 1) have also been correlated from geochemical studies with Eskimo and Persillon volcanics (Legault et al. 1994), suggesting that they may belong to the Minto dyke swarm. The Minto, Eskimo, and Persillon paleopoles are quite different from the prefold paleopoles of the undated Flaherty volcanics and Haig intrusions (Fig. 11*b*) that overlie the Eskimo volcanics

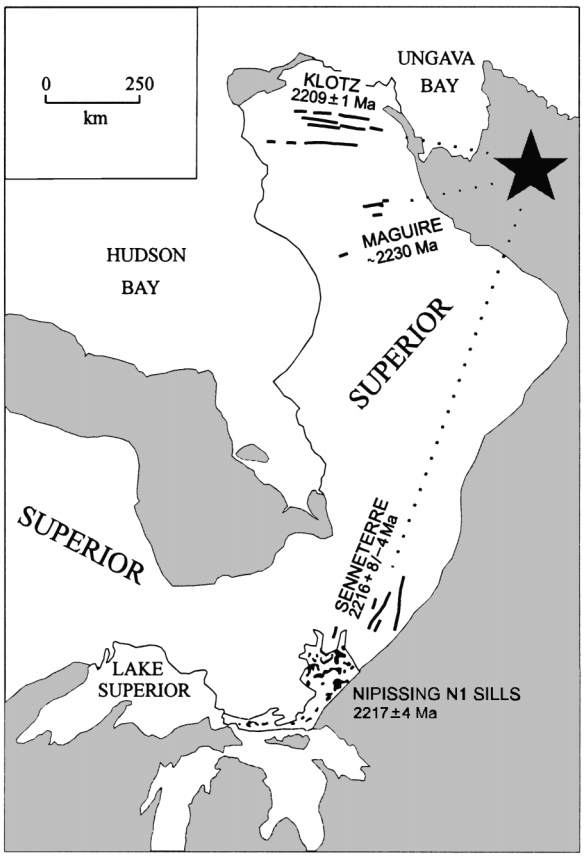
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Fig. 11. Comparison of paleomagnetic poles for Ungava dykes of this study with (*a*) other ca. 2.5–2.0 Ga primary paleopoles from the Superior and Southern provinces and (*b*) post-2.0 Ga paleopoles from the Circum-Ungava fold belt. Ovals of 95% confidence about the mean poles are shown. The dashed oval for the Persillon North pole indicates that this pole is plotted in the opposite hemisphere at 49°N, 134°E. The poles for the Eskimo volcanics, and the overlying Flaherty volcanics and the Haig intrusions of the Circum-Ungava fold belt, are prefold. Only U–Pb ages are indicated. Large arrows indicate the trend in ages of paleopoles; they do not represent an apparent polar wander path. Matachewan dyke poles and age from Bates and Halls (1990) and Heaman (1997), respectively; Senneterre dyke pole and age from Buchan et al. (1993, 1996); Nipissing N1 sills grand mean pole calculated herein from poles reported in Symons (1970, 1971), Patel and Palmer (1974), Symons and Londry (1975), Roy and Lapointe (1976), and Stupavsky and Symons (1982), and Nipissing N1 age from Noble and Lightfoot (1992); Marathon dyke poles and age from Buchan et al. (1996); Fort Frances dyke pole and age from Halls (1986) and Buchan et al. (1996), respectively; Eskimo volcanic, Flaherty volcanic, and Haig intrusion poles from Schmidt (1980); Persillon North and South volcanic poles (for volcanics north and south of Richmond Gulf) from Schwarz and Fujiwara (1981). Ptarmigan virtual geomagnetic pole, based on only three sampling sites, does not average out secular variation.



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Fig. 12. Proposed giant radiating dyke swarm consisting of Klotz and Maguire dykes of the Minto block and Senneterre dykes of the southern Superior Province. The swarm is centred southeast of Ungava Bay (star) and may have fed coeval Nipissing N1 sills of the Southern Province.



on the Belcher Islands. Therefore, the Flaherty and Haig units are likely younger than 1998 Ma.

Key paleopoles with high-precision ages (Buchan et al. 1994) are a prerequisite for tracking the drift of the Superior Province and reconstructing the continents of which it was a part in the Paleoproterozoic. The key paleopoles of Fig. 11*a* locate the Superior Province in low latitudes (10–20°) between ca. 2.50 and 2.45 Ga and again at ca. 2.22 Ga. Between 2.22 and 2.17 Ga the craton rotated counterclockwise through an angle of approximately 90° while drifting to intermediate latitudes of 30–40°. Between 2.17 and 2.00 Ga a further counterclockwise rotation of about 50° is recorded by the paleomagnetic data as the craton drifted back to lower latitudes of 20–30°. Key Paleoproterozoic paleopoles of equivalent ages are not yet available from other cratons. Therefore, continental reconstructions incorporating the Superior Province are not yet possible.

As noted above, comparison of paleopoles from Maguire dykes of the Minto block and similar-aged Senneterre dykes of the southern Superior Province and Nipissing sills of the Southern Province indicates no measurable relative rotation between the Minto block and southern Superior Province since the emplacement of these units at ca. 2.22 Ga. Thus, the relative trends of the Senneterre, Maguire, and Klotz dykes are primary. They form a roughly radial pattern (Fig. 12) with a focal point in the region southeast of Ungava Bay. We propose that the three dyke sets may be subswarms of a giant radiating dyke swarm whose focus is located above the site of a ca. 2.22 Ga mantle plume. If so, the Senneterre dykes, which are not traceable on current aeromagnetic maps, should extend northeastward from where they have been identified in the Senneterre area by geological mapping and paleomagnetism (Fig. 12) towards the proposed plume centre. In particular, two north-northeast- trending dykes of the Lacs Obatogamau area about 230 km northeast of Senneterre, from which Senneterre remanence directions have been obtained (W.F. Fahrig, personal communication in Buchan et al. 1993), could represent an extension of the Senneterre swarm.

About 30 giant radiating swarms have been identified on Earth (e.g., Ernst and Buchan 1997*a*) and more than 110 are known on Venus (e.g., Ernst et al. 1995). Dramatic examples on Earth include the 1.27 Ga Mackenzie swarm of northern Canada (Fahrig and Jones 1969; LeCheminant and Heaman 1989), which extends for 2600 km from its focus, and the 0.20 Ga Central Atlantic reconstructed swarm (May 1971) of eastern North America, northern South America, and northwestern Africa, which extends for 2800 km from its focus. Theoretical modelling (e.g., Rubin 1995) indicates that even narrow dykes a few metres thick can propagate for such distances. Magnetic fabric studies demonstrate that beyond a distance of a few hundred kilometres, which may correspond to the outer boundary of subcrustal melt generation in the mantle plume, magma flow in giant radiating dyke swarms is subhorizontal (Ernst and Baragar 1992; Greenough and Hodych 1990).

In many examples the plume centre associated with giant radiating dyke swarms is located along a continental margin, suggesting that the plume event was responsible for continental breakup (Fahrig 1987; Ernst and Buchan 1997*a*). The postulated ca. 2.22 Ga Ungava plume centre, related to the Senneterre, Maguire, and Klotz swarms, lies off the northeastern margin of the Superior Province (Fig. 12) and may be related to continental breakup along this margin in the Paleoproterozoic. If so, then the rest of the swarm is located on a presently unidentified continental block or blocks that rifted away, in direct analogy with the Central Atlantic swarm cited above.

The 2217–2210 Ma Nipissing sills of the Southern Province (Fig. 12) are an extensive set of intrusions in the sedimentary units of the Paleoproterozoic Huronian Supergroup along the southern margin of the Superior Province. As noted earlier, they are indistinguishable in age from the dykes of the Senneterre, Maguire, and Klotz swarms. In addition, the dominant N1 magnetization direction of Nipissing sills is similar to the magnetization direction of Senneterre and Maguire dykes (Fig. 11*a*). We propose that the sills represent a portion of the Ungava magmatic event and that the Nipissing sills were fed laterally from the Ungava plume centre via the Senneterre dykes. In this scenario, the magma flowed horizontally in Senneterre dykes until encountering the sedimentary strata of the Huronian Supergroup about 1500 km from the proposed Ungava plume centre (Fig. 12), where magma from some of the dykes intruded as Nipissing sills. Fahrig (1987) earlier proposed that Nipissing sills might be fed from the northeast, although dating of the

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dykes and sills was not then sufficiently precise to identify the feeders. Other examples in which sill complexes are thought to be fed from great distances by giant radiating dyke swarms (Ernst and Buchan 1997*b*) include the Christie Bay sills of Great Slave Lake and sills in the Athabasca Basin, located 900 and 1400 km, respectively, from the focus and presumed mantle plume centre of the 1.27 Ga Mackenzie dyke swarm, and the Coronation sills, about 800 km from the focus of the 0.72 Ga Franklin swarm.

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